

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Fifth Semester of B. Tech (CL) Examination****March-April 2018****CL303 Geotechnical Engineering - I****Date: 31.03.2018, Saturday****Time: 01:30 p.m. To 04:30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 (a)** Define: loam & peat. **[02]****(b)** What is seepage velocity? How it is related to discharge velocity? **[03]****(c)** Write a brief note on scope of soil engineering. **[04]****Q - 2 (a)** 1 m³ of wet soil weigh 19.8 kN. If the specific gravity of soil particles is 2.7 & water content is 11 %, find void ratio, dry density and degree of saturation. **[06]****(b)** An undisturbed specimen of clay was tested in laboratory & following results were obtained. **[06]**

Wet mass = 210 gm

Oven dry mass = 175 gm

Specific Gravity = 2.7

What was the total volume of the original undisturbed specimen assuming that the specimen was 50 % saturated?

OR**(b)** Enlist various methods to find specific gravity of soil and explain pycnometer method in brief. **[06]****Q - 3 (a)** 400 gm dry soil was used for sieve analysis. The mass of soil retained on each sieve are given below. Plot a grain size distribution curve and classify the soil. **[10]**

Sieve analysis data								
IS sieve size (mm)	4.75	2.0	1.0	0.425	0.212	0.150	0.075	Pan
Wt. retained on sieve(gm)	16.80	50	150	106	46.90	16.20	8.70	5.40

(b) Explain dispersed structure with neat sketch. **[04]****OR****(b)** Classify the soil with the following properties according to IS classification system. **[04]**

	% passing 4.75 mm sieve	% passing 75 μ sieve	Liquid limit (%)	Plasticity Index (%)
Soil - A	65	40	30	15

SECTION – II

- Q - 4 (a)** Highlight effect of water content & compactive energy on compaction of soil. [03]
- (b)** Distinguish between Standard Proctor Test and Modified Proctor Test. [05]
- (c)** How to find permeability of soil by falling head permeability test? [05]
- Q - 5 (a)** Write a short note on unconfined compression test. [05]
- (b)** A cylindrical specimen of saturated clay, 4 cm in diameter and 8.66 cm in overall length is tested in unconfined compression test apparatus. Find the unconfined compressive strength of a clay, if the specimen fails under an axial load of 46.5 N. The change in length of specimen at failure is 1 cm. [06]

OR

- (b)** A vane, 10 cm long and 8 cm in diameter, was pressed into soft clay at the bottom of bore hole. Torque was applied and gradually increased to 45 N-m when failure took place. Subsequently, the vane rotated rapidly so as to completely remold the soil. The remolded soil was sheared at a torque of 18 N-m. Calculate the cohesion of clay in natural and remolded states and also the value of sensitivity. [06]
- Q - 6 (a)** Explain forward method of finding void ratio from consolidation test data. [05]

OR

- (a)** Explain concept of consolidation with spring analogy. [05]
- (b)** Two clay specimen A and B, of thickness 2 cm & 3 cm, have equilibrium void ratios 0.68 and 0.72 respectively under a pressure of 200 kN/m^2 . If the equilibrium void ratios of the two soils reduced to 0.50 and 0.62 respectively, when the pressure was increased to 400 kN/m^2 , find the ratio of permeability of two soil specimens. Time required by A specimen to reach 40 % degree of consolidation is one fourth of that required by B specimen to reach 40 % degree of consolidation. [06]
